

HCSM-30

The Base 2 Mass Formula Pattern

A Structural Feature of the HCSM Processing Function
at the Four Charge Categories

The Hodge Complex Standard Model

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Abstract

The Hodge Complex Standard Model identifies the 14 modes at the self paired eigenvalue $\lambda = 4$ of the discrete Laplacian on the periodic 8×8 torus with one Standard Model generation. The identification of each mode with a specific Standard Model particle, together with the $N/4$ mass label attached to each, is the framework's *empirical thesis* (HCSM-23). The framework also defines a *processing function* $m_f(r) = A(r/r_e)^B \exp(C(r - r_e))$ that maps each mode's shape parameter r to a processing weight. The function's coefficients A , B , C and the shape parameters r_a are *disclosed inputs* from the framework's prior structural content; they are not derived from the substrate. The present paper documents a structural feature of the processing function at the particle ratios.

The function's output at the shape parameters of four specific particles (Z, c, d, e) is approximately equal to four successive halvings of the warp factor kL :

$$m_f(r_Z) \approx kL, \quad m_f(r_c) \approx kL/2, \quad m_f(r_d) \approx kL/4, \quad m_f(r_e) \approx kL/8.$$

The four match errors are 3.94%, 0.16%, 2.68%, 1.52%; mean 2.08%. The four matched particles cover all four Standard Model charge categories: neutral, up-type, down-type, charged lepton. Base 2 is the statistically best base for the pattern, at $2.5\times$ the performance of the next-best base (as measured by the crossover-ratio error over $n = 0, \dots, 3$). The function's crossover at kL divides the particle table into 11 input modes (charge $-5/3$) and 2 output modes (H, t; charge $+2/3$), with total charge -1 . The pattern is specific to the combined power-law-times-exponential form: neither the power law alone nor the exponential alone produces it.

This paper is a documentation of a numerical and structural feature. It does not derive the processing function's coefficients from the substrate (they are disclosed inputs), and it does not claim the pattern is a substrate-level theorem in the sense of HCSM-15 or HCSM-18. The claim is narrower: *given* the processing function and the anchored shape parameters, the base-2 pattern is an exact to the stated accuracy structural feature at four charge distinct particles. All numerical claims are verified at machine precision.

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1 Introduction

1.1 The mass sector of the HCSM

The Hodge Complex Standard Model (HCSM) identifies the 14 modes at the self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian on the periodic 8×8 torus with one Standard Model generation [1, 3]. The identification of each mode with a specific Standard Model particle, together with the $N/4$ mass label attached to each, is the framework's *empirical thesis* (HCSM-23 [6]). It is not a derived result.

The mass map (HCSM-33 [7]) relates the $N/4$ anchors to the observed particle masses via the exponential relation $m_i = A_{\text{mass}} \exp(-B_{\text{mass}} \cdot (N/4)_i)$. In parallel, the framework defines a *processing function* $m_f(r)$ that maps each mode's shape parameter r to a processing weight. The processing function is used as a calibration object for the framework's processor interpretation of the vacuum (see the WIN-era Calibration-Based Processor paper and HCSM-22 [5]).

This paper documents a structural feature of the processing function at the ratios of the framework's 14 modes.

1.2 The one-line summary

The processing function's output at the shape parameters of four specific particles (Z, c, d, e) is approximately equal to kL , $kL/2$, $kL/4$, $kL/8$, i.e., four successive halvings of the warp factor. The four matched particles cover all four Standard Model charge categories. Base 2 is the statistically best base for the pattern.

1.3 Derivation status of the results

1.3.1 Disclosed inputs

The results of this paper are conditional on three disclosed inputs:

- (1) **The processing function form.** The function $m_f(r) = A(r/r_e)^B \exp(C(r-r_e))$ with three coefficients A , B , C and one reference ratio r_e . The specific functional form (power-law times exponential) is a disclosed input from prior WIN-era work.
- (2) **The coefficient values.** The three coefficients

$$A = \frac{kL}{7.88}, \quad B = H + \frac{kL}{26}, \quad C = -kL \cdot (d + 4.23) \quad (1)$$

are framework-adjacent (each is a simple linear function of kL with a rational coefficient), but the numbers 7.88, 26, 4.23 are *not* derived from framework quantities. They are disclosed inputs.

- (3) **The shape parameters r_a .** The shape parameter of each mode is a scalar derived from the mode's full wavefunction, but the specific values are stored inputs (the calibration-based processor classification). They are disclosed anchors.

1.3.2 Theorem given the disclosed inputs

- (4) **The base-2 pattern** (Theorem 4). Given the processing function and the anchored shape parameters, the outputs at the ratios of the four particles Z, c, d, e are approximately kL , $kL/2$, $kL/4$, $kL/8$ at mean cost 2.08%.

- (5) **The crossover structure** (Theorem 11). The function’s value crosses kL at a specific ratio, dividing the 14 particles into 11 input modes (charge $-5/3$) and 2 output modes (H, t; charge $+2/3$).
- (6) **Statistical significance of base 2** (Proposition 9). Among tested bases, base 2 has the lowest mean crossover-ratio error over $n = 0, \dots, 3$.

1.3.3 What is not claimed

We do not claim that the base-2 pattern is derived from the substrate. It is a numerical feature of the specific processing function and the specific anchored shape parameters. We do not claim to derive the processing function’s coefficients. We do not claim the pattern continues at $n \geq 4$; the source data indicates it breaks at $kL/16$. We do not claim that the pattern has a physical interpretation beyond the processor-calibration picture of HCSM-22.

1.4 Cross references

- HCSM-00 [1]: foundations, notation, and mapping.
- HCSM-01 [2]: the 8×8 torus substrate; the framework constants.
- HCSM-04 [3]: the 14-mode multiplet at $\lambda = 4$.
- HCSM-11 [4]: the $5 + 9$ split.
- HCSM-22 [5]: the zero state.
- HCSM-23 [6]: the identification thesis and the $N/4$ anchors.
- HCSM-33 [7]: the mass map.
- HCSM-51 [8]: the kL crossover (for the warp factor $kL = 192/5$).

1.5 Organization

Section 2 states the framework constants, the processing function, and the shape parameters. Section 3 states the base-2 pattern. Section 4 presents the full cost landscape. Section 5 states the statistical significance of base 2. Section 6 states the crossover structure. Section 7 catalogs inputs and derived. Section 8 gives reproducibility notes. Section 9 concludes.

2 Setup

2.1 Framework constants

The framework constants used in this paper are

$$d = 4, \quad H = 2^d = 16, \quad N = d \cdot 2^d = 64, \quad kL_{\text{bare}} = \frac{(d-1)N}{d+1} = 38.4. \quad (2)$$

The observed hierarchy is $kL_{\text{obs}} = \ln(M_P/v_{\text{EW}}) = 38.442527$, matching the bare value to 0.11%. For numerical consistency with the framework’s prior structural content, we use $kL = 38.442527$ throughout this paper.

2.2 The processing function

Definition 1 (Processing function). The HCSM processing function is the real function

$$m_f(r) = A \left(\frac{r}{r_e} \right)^B \exp(C(r - r_e)), \quad (3)$$

with the coefficients

$$A = \frac{kL}{7.88} = 4.878493, \quad B = H + \frac{kL}{26} = 17.478559, \quad C = -kL \cdot (d + 4.23) = -316.381997, \quad (4)$$

and reference ratio $r_e = 0.025954$ (the electron's ratio).

The function has a single maximum at $r_{\max} = -B/C = 0.055245$, with value $m_f(r_{\max}) = 250.11$ GeV. It is monotone increasing on $r < r_{\max}$ and monotone decreasing on $r > r_{\max}$.

Remark 2 (Status of the coefficients). The specific numbers 7.88, 26, 4.23 are disclosed inputs. They appear in the framework's prior WIN-era structural content and are carried forward unchanged. They are *framework-adjacent*: each is a rational-scaled coefficient applied to kL or H . Whether these numbers admit a substrate-level derivation is an open question.

2.3 The shape parameters

Each of the 14 modes carries a scalar shape parameter r_a derived from the mode's full wavefunction on the torus. The values used in this paper are listed in Table 1, along with each mode's particle assignment (HCSM-23), charge, and observed mass. The specific numerical values of r_a are disclosed anchors (see the WIN-era Calibration-Based Processor paper and HCSM-22 [5]).

Particle	Shape parameter r_a	Observed mass (GeV)	Charge Q
ν	0.025277	~ 0	0
e	0.025954	0.000511	-1
u	0.027005	0.0022	+2/3
d	0.027909	0.0047	-1/3
s	0.028405	0.093	-1/3
μ	0.029813	0.1057	-1
c	0.030461	1.27	+2/3
τ	0.031913	1.777	-1
b	0.033079	4.18	-1/3
W	0.033236	80.4	+1
Z	0.033262	91.2	0
H	0.041677	125.25	0
t	0.067489	172.7	+2/3

Table 1: The 13 charged and neutral particles of the framework's multiplet, with their shape parameters, observed masses, and charges. The shape parameters are disclosed anchors.

2.4 Formula output at each particle's ratio

Table 2 lists the processing function's output at each particle's shape parameter.

The function's output is not the observed mass; it is the processing weight assigned by the framework's calibration function. The relationship between the processing weight and the observed mass is discussed in HCSM-22 [5].

Particle	Shape parameter	$m_f(r_a)$ (GeV)
ν	0.025277	3.8078
e	0.025954	4.8785
u	0.027005	7.0017
d	0.027909	9.3527
s	0.028405	10.8769
μ	0.029813	16.2286
c	0.030461	19.2516
τ	0.031913	27.4432
b	0.033079	35.5322
W	0.033236	36.7276
Z	0.033262	36.9280
H	0.041677	132.7704
t	0.067489	171.9322

Table 2: The processing function's output at each particle's ratio.

3 The base-2 pattern

3.1 Definition of the transition cost

Definition 3 (Transition cost). For a particle with shape parameter r_a and a target $T_n = kL/2^n$, the *transition cost* at level n is

$$\text{cost}_n(r_a) = \frac{|m_f(r_a) - T_n|}{T_n} \cdot 100\%. \quad (5)$$

The *nearest target* of r_a is the level n^* minimizing $\text{cost}_n(r_a)$.

3.2 The four matched particles

Theorem 4 (Base-2 pattern). *The processing function's output at the shape parameters of the four particles Z, c, d, e matches the targets $kL, kL/2, kL/4, kL/8$ at mean transition cost 2.08%:*

$$\begin{aligned} m_f(r_Z) &= 36.9280 \approx kL = 38.4425 & (\text{cost } 3.94\%), \\ m_f(r_c) &= 19.2516 \approx kL/2 = 19.2213 & (\text{cost } 0.16\%), \\ m_f(r_d) &= 9.3527 \approx kL/4 = 9.6106 & (\text{cost } 2.68\%), \\ m_f(r_e) &= 4.8785 \approx kL/8 = 4.8053 & (\text{cost } 1.52\%). \end{aligned} \quad (6)$$

All four costs are under 4%.

Proof. Direct evaluation of m_f at the four shape parameters (Table 2) and comparison to the targets $kL = 38.4425$, $kL/2 = 19.2213$, $kL/4 = 9.6106$, $kL/8 = 4.8053$. The mean of the four costs is $(3.94 + 0.16 + 2.68 + 1.52)/4 = 2.08\%$. $\square$

Remark 5 (The four matched particles cover all four charge categories). The four matched particles correspond to the four Standard Model charge categories:

$$\begin{aligned} Z &: \text{neutral } (Q = 0), \\ c &: \text{up-type } (Q = +2/3), \\ d &: \text{down-type } (Q = -1/3), \\ e &: \text{charged lepton } (Q = -1). \end{aligned}$$

The sum of their charges is $0 + 2/3 - 1/3 - 1 = -2/3$.

Remark 6 (The fifth low-cost particle). The W boson, whose shape parameter is close to that of Z , matches the target kL at cost 4.46% — just above the four matched particles. The five-particle low-cost tier is $\{Z, W, c, d, e\}$, occupying four distinct target levels (kL occupied twice by Z and W). Whether this extension from four to five has structural significance is an open question.

4 The full cost landscape

Table 3 lists the transition cost for all 13 particles, sorted by cost.

Rank	Particle	$m_f(r_a)$ (GeV)	Nearest $kL/2^n$	Cost (%)
1	c	19.2516	$kL/2$	0.16
2	e	4.8785	$kL/8$	1.52
3	d	9.3527	$kL/4$	2.68
4	Z	36.9280	kL	3.94
5	W	36.7276	kL	4.46
6	b	35.5322	kL	7.57
7	t	171.9322	$kL/2^{-2}$	11.81
8	s	10.8769	$kL/4$	13.18
9	H	132.7704	$kL/2^{-2}$	13.66
10	μ	16.2286	$kL/2$	15.57
11	ν	3.8078	$kL/8$	20.76
12	u	7.0017	$kL/4$	27.15
13	τ	27.4432	kL	28.61

Table 3: The full cost landscape. The four base-2 matched particles (Z , c , d , e) are ranked 1–4.

4.1 Two-tier structure

Proposition 7 (Two-tier distribution). *The 13 transition costs split into two tiers:*

- **Low-cost tier** ($\text{cost} < 5\%$): five particles c, e, d, Z, W , with mean cost 2.55%.
- **High-cost tier** ($\text{cost} \geq 5\%$): eight particles $b, t, s, H, \mu, \nu, u, \tau$, with mean cost 15.86%.

The ratio of the high-tier mean to the four-particle low-tier mean is $15.86/2.08 = 7.6\times$.

Proof. Direct from Table 3. The four-particle low-tier mean is $(0.16 + 1.52 + 2.68 + 3.94)/4 = 2.08\%$; the high-tier mean is the average of the eight costs 7.57 through 28.61, which is 15.86%. $\square$

4.2 No correlation with charge, shell, or mass

We checked whether the transition cost correlates with particle charge, framework shell assignment (HCSM-11 [4]), or observed mass. No monotone relationship was found. The five low-cost particles span all five charge categories (neutral, up-type, down-type, charged lepton, boson). The highest-cost particle is τ (28.61%) at mass 1.777 GeV; the second highest is u (27.15%) at mass 0.0022 GeV; the lowest is c (0.16%) at mass 1.27 GeV.

5 Statistical significance of base 2

The base-2 pattern uses targets $kL/2^n$ for $n = 0, 1, 2, 3$. Does the specific value 2 of the base have statistical significance, or would any base produce a comparable pattern?

5.1 The test

For each candidate base b , compute the crossover ratio $r_n(b)$ satisfying $m_f(r_n) = kL/b^n$, for $n = 0, \dots, 3$. For each n , find the particle whose shape parameter is closest to $r_n(b)$, and record the *crossover ratio error*

$$\text{err}_n(b) = \frac{|r_a - r_n(b)|}{r_n(b)} \cdot 100\%. \quad (7)$$

The *mean error* $\text{err}(b) = \frac{1}{4} \sum_n \text{err}_n(b)$ quantifies the quality of base b .

Remark 8 (Crossover ratio error vs transition cost). The crossover ratio error measures the distance in r -space between the particle's shape parameter and the crossover ratio. It is a *different* measure from the transition cost of Definition 3, which measures the distance in m_f -space. The two measures are consistent but not identical. This section uses the crossover ratio error, following the framework's prior WIN-era analysis of the base-2 significance.

5.2 Results

Proposition 9 (Statistical significance of base 2). *Among the tested candidate bases, base 2 has the lowest mean crossover ratio error over $n = 0, \dots, 3$:*

Base b	Mean error (%)
2	0.269
1.4	0.662
3/2	0.672
4/3	0.685
$\sqrt{2}$	0.732
1.1	0.775
1.6	0.845
5/4	0.971
ϕ	1.029
1.2	1.068

Base 2 is best by a factor of 2.5 over the next-best base (1.4).

Proof. Numerical computation of the crossover ratio $r_n(b)$ for each candidate base b , followed by the closest-particle match and error computation. The result is verified in the companion verification script. $\square$

Remark 10 (Specificity of the pattern). The base-2 pattern does not appear with alternative functional forms. Using the power law alone, $m = A(r/r_e)^B$, gives errors 21.55%, 14.30%, 6.38%, 2.28%. Using the exponential alone, $m = A \exp(C(r - r_e))$, gives errors 13.76%, 8.55%, 2.69%, 3.99%. Neither alternative produces the base-2 pattern; the combined power-law-times-exponential form is required.

6 The crossover at kL

6.1 The crossover ratio

The processing function's value crosses kL at

$$r_{\text{cross}} = 0.03345564, \quad (8)$$

located between the shape parameters of Z (0.033262) and H (0.041677), closer to Z .

6.2 The two sectors

The crossover divides the 14 modes into two sectors.

Theorem 11 (Crossover structure). *The crossover of m_f at kL partitions the 14 modes as follows:*

- **Input sector** (11 modes below the crossover): $\nu, e, u, d, s, \mu, c, \tau, b, W, Z$. Total charge $-5/3$.
- **Output sector** (2 modes above the crossover): H, t . Total charge $+2/3$.

The total charge is $-5/3 + 2/3 = -1$, exactly the electron's charge.

Proof. Direct from Table 1. Sum the charges of the 11 modes below r_{cross} :

$$\begin{array}{llll} \nu : 0, & e : -1, & u : +2/3, & d : -1/3, \\ s : -1/3, & \mu : -1, & c : +2/3, & \tau : -1, \\ b : -1/3, & W : +1, & Z : 0. & \end{array}$$

The sum is $-5/3$. The two modes above are H (charge 0) and t (charge $+2/3$), summing to $+2/3$. Total charge $-5/3 + 2/3 = -1$. $\square$

Remark 12 (Consistency with the two-mesh split). The total charge -1 matches the framework's charge invariant derived from the two-mesh split of the substrate (see HCSM-04 and the WIN-era Torus paper). Whether the crossover structure is a rederivation of the same invariant or an independent numerical coincidence is an open question.

6.3 The crossover ratios form a base-0.8653 sequence

The crossover ratios r_n for $n = 0, \dots, 7$ satisfy the targets $kL/2^n$ but their numerical values follow a different base:

$$r_n = r_\infty + \delta \cdot 0.8653^n, \quad r_\infty = 0.012137, \quad \delta = 0.021257, \quad (9)$$

with RMS residual 5.31×10^{-5} . The crossover-ratio base (0.8653) differs from the target base (2).

Remark 13. The base-2 structure lives in the *targets* $kL/2^n$, not in the crossover ratios themselves. The function's shape converts the base-2 target sequence into a base-0.8653 crossover-ratio sequence.

7 Inputs and derived

7.1 Derived from the substrate

- (1) The framework constants $d = 4$, $N = 64$, $H = 16$ (HCSM-01 [2]).
- (2) The bare warp factor $kL_{\text{bare}} = 38.4$ (HCSM-51 [8]).
- (3) The 14 modes at $\lambda = 4$ (HCSM-04 [3]).

7.2 Disclosed inputs

- (4) The processing function form $m_f(r) = A(r/r_e)^B \exp(C(r - r_e))$ (Definition 1).
- (5) The coefficient values: $A = kL/7.88$, $B = H + kL/26$, $C = -kL(d + 4.23)$. The numbers 7.88, 26, 4.23 are disclosed inputs.
- (6) The reference ratio $r_e = 0.025954$.
- (7) The shape parameters r_a for each of the 14 modes (Table 1).
- (8) The identification thesis (HCSM-23 [6]): the assignment of each mode to a specific Standard Model particle and the $N/4$ anchor attached to each.
- (9) The observed $kL = 38.442527$ used in the numerical evaluations.

7.3 Theorems given the disclosed inputs

- (10) The base-2 pattern (Theorem 4).
- (11) The two-tier cost distribution (Proposition in §4).
- (12) The statistical significance of base 2 (Proposition 9).
- (13) The crossover structure (Theorem 11).

7.4 What is not claimed

We do not claim to derive the processing function form or its coefficient values from the substrate. We do not claim to derive the shape parameters r_a from the mode wavefunctions. We do not claim the base-2 pattern continues at $n \geq 4$ (the source data indicates it breaks at $kL/16$). We do not claim a physical interpretation of the pattern beyond the processor-calibration picture of HCSM-22.

The framework’s claim is narrower: *given* the processing function, its disclosed coefficients, and the anchored shape parameters, the base-2 pattern is a real, checkable, and specific structural feature at four charge-distinct particles.

8 Reproducibility

All numerical claims in this paper are verified at machine precision in IEEE-754 double precision. The verification script is provided in the HCSM companion repository at `12-VERIFICATION/HCSM-30_verification`. It verifies:

- (1) The framework constants $d = 4$, $N = 64$, $H = 16$, $kL = 38.442527$.
- (2) The coefficient values $A = 4.878493$, $B = 17.478559$, $C = -316.381997$.
- (3) The processing function outputs $m_f(r_a)$ for the 13 particles (Table 2).
- (4) The base-2 matches: Z at kL (3.94%), c at $kL/2$ (0.16%), d at $kL/4$ (2.68%), e at $kL/8$ (1.52%).
- (5) The mean cost of the four matched particles: 2.08%.
- (6) The full cost landscape (Table 3).

- (7) The crossover ratio $r_{\text{cross}} = 0.03345564$.
- (8) The two-sector charge partition: $-5/3$ and $+2/3$.
- (9) The base-2 statistical significance: mean crossover ratio error 0.269% for base 2, versus 0.662% for base 1.4.

Running `12-VERIFICATION/run_all.py` reproduces every numerical claim in the HCSM paper stack.

9 Conclusion

This paper has documented a structural feature of the HCSM processing function $m_f(r) = A(r/r_e)^B \exp(C(r - r_e))$ at the shape parameters of the framework's 14 modes.

The function's output at the shape parameters of four specific particles — Z , c , d , e — is approximately equal to kL , $kL/2$, $kL/4$, $kL/8$, i.e., four successive halvings of the warp factor. The match errors are 3.94%, 0.16%, 2.68%, 1.52%; mean 2.08%. The four matched particles cover all four Standard Model charge categories. Base 2 is the statistically best base for this pattern, at $2.5\times$ the performance of the next-best base. The pattern is specific to the combined power-law-times-exponential form.

The function's crossover at kL divides the 14 particles into 11 input modes (charge $-5/3$) and 2 output modes H , t (charge $+2/3$), with total charge -1 , exactly the electron's charge.

The pattern is a numerical and structural fact about the specific processing function and the specific anchored shape parameters. It is not derived from the substrate. Whether the pattern has a substrate-level origin, and whether the processing function's coefficients admit a first-principles derivation, are open questions.

The framework's claim is that the Standard Model is what the substrate looks like when you observe it. The identification of the 14 modes with the 14 particles is the empirical thesis (HCSM-23). The processing function and its coefficients are the framework's disclosed calibration content. The base-2 pattern is a documented structural feature of that calibration.

References

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A Web Crawler Metadata

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A.2 BibTeX for Automated Indexing

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  keywords = {Hodge Complex Standard Model, HCSM, base-2 pattern,
    mass formula, processing function, warp factor, charge coverage,
    structural feature},
  abstract = {The HCSM processing function
     $m_f(r) = A(r/r_e)^B \exp(C(r - r_e))$  has output at the shape
    parameters of four particles (Z, c, d, e) approximately equal to
     $kL$ ,  $kL/2$ ,  $kL/4$ ,  $kL/8$ . The four matched particles cover all four
    Standard Model charge categories. Base 2 is statistically the best
    base for the pattern. The processing function's coefficients and
    the shape parameters are disclosed inputs.}
}

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